Fakulti:	FAKULTI KEJURUTERAAN ELEKTRIK
Nama Matapelajaran: Makmal Umum Tahun 3 Kod Matapelajaran : SKEE 3732	Semakan : 1 Tarikh Keluaran : October 2025 Pindaan Terakhir : No. Prosedur : **PK-UTM-SKE-(0)-10**

FAKULTI KEJURUTERAAN ELEKTRIK
UNIVERSITI TEKNOLOGI MALAYSIA
KAMPUS SKUDAI
JOHOR

SKEE 3732

MICROPROCESSOR LABORATORY

Laboratory 2: GPIO and Interrupt Programming

Prepared by:	Approved by : Director of Programme
Dr Mohd Afzan Othman	Name: Prof. Ir Dr Rubita Sudirman
Signature & Stamp: DR. MOHD AFZAN BIN OTHMAN Pensyarah Kanan Jabatan Elektronik dan Kejuruteraan Komputer Fakulti Kejuruteraan Elektrik Universiti Teknologi Malaysia 81310 Johor Bahru Johor, Malaysia	Signature & Stamp: PROFESOR IR. DR. RUBITA BINTI SUDIRMAN Head of Department Department of Electronic and Computer Engineering Faculty of Electrical Engineering Universiti Teknologi Malaysia 81310 UTM Johor Bahru, Johor
Date: 2 October 2025	Date: 2 October 2025

I. PRELIMINARY PREPARATION (15 marks)

Important Note: You are required to do following BEFORE the lab session.

1. Explain the function of pull-up and pull-down resistors in push button circuits. Why are they important?
2. Identify the register in STM32 used to:
   - Configure pin direction (input/output).
   - Enable internal pull-up or pull-down resistors.
   - Read pin input value.
   - Write output value to a pin.
3. Using a suitable software (e.g., Proteus, Multisim, KiCad or EasyEDA), draw a complete circuit schematic based on the description in Table 1. Present the circuit to your supervisor for verification before proceeding with Lab 2.

Table 1: STM32 Connection

Device	STM32 pin	GPIO Port	Remarks
Button0	PA0	GPIOA	Active HIGH
Button1	PA1	GPIOA	Active HIGH
Button2	PA2	GPIOA	Active HIGH
7-segment Display1	PB0-PB7	GPIOB	Common Cathode
7-segment Display2	PC0-PC7	GPIOC	Common Cathode

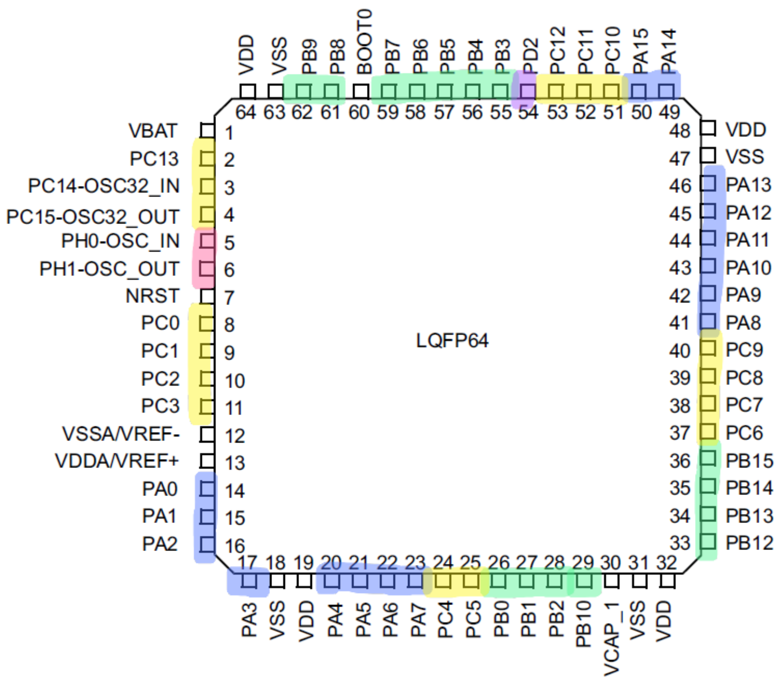

Figure 1: STM32F446RE Pin Out

II. LABORATORY SHEET

1 Title: Voting System Implementation using STM32 Nucleo-64 (F446RE)

2 Objective:

At the end of this laboratory session, students will be able to:

1. To understand and apply bare-metal C programming on ARM Cortex-M4 STM32 microcontroller.
2. To configure GPIO pins for input and output.
3. To design and implement a simple voting system that counts votes for two candidates.
4. To use a reset button to clear voting results.

3 Equipment/Software/Reference:

1. STM32 Nucleo-64 F446RE development board
2. Breadboard and jumper wires
3. 3 push button switches with pull-down resistors (1 kΩ)
4. 2 seven-segment display (common cathode) with resistors (330 Ω)
5. USB cable for programming/debugging
6. STM32CubeIDE (for writing and flashing code)

4 Background Theory

The STM32F446RE is a Cortex-M4 based ARM microcontroller capable of handling real-time embedded applications. In this lab, students will develop a voting system that allows users to press buttons to vote for two candidates. Each candidate’s vote count will be displayed on a 7-segment display. A third button will reset the vote count for both candidates.

The system requires GPIO input for buttons and GPIO output for driving the 7-segment displays. Debouncing techniques may be necessary to ensure accurate button presses. Students will practice bare-metal programming by directly manipulating STM32 registers (RCC, GPIOx, etc.) instead of using HAL libraries.

5 Procedures

Note: You must complete Preliminary Preparation before proceeding this section. Make sure that STM32CubeIDE has been installed on your PC or Laptop.

You are required to develop a voting system that allows users to press buttons to vote for two candidates. Each candidate's vote count will be displayed on a 7-segment display. A third button will reset the vote count for both candidates.

<u>Push Button Connections</u>

- Button 2 → Candidate A (increment A's count)
- Button 1 → Candidate B (increment B's count)
- Button 0 → Reset votes (reset both counts to 0)

<u>7-Segment Display Connections</u>

- Display 1 → Candidate A's votes.
- Display 2 → Candidate B's votes.

<u>Board Setup</u>

- Connect the STM32 Nucleo board via USB to PC.
- Load program using STM32CubeIDE.

Procedures

1. Write a complete bare-metal C program for the voting system above by following steps below:
    i. Initialize system clock and enable GPIO ports using RCC registers.
    ii. Configure button pins as input with pull-down resistors.
    iii. Configure 7-segment pins as output.
    iv. Write functions to:
        - Detect button press.
        - Increment/decrement vote counters.
        - Reset vote counters.
        - Display numbers (0–9) on 7-segment.
2. Connect the STM32 Nucleo-64 board with all components listed in Table 1, based on the circuit designed during the pre-lab session. Refer Figure 2 for the pins out connection. Present the completed circuit to the supervising lecturer for verification before uploading the program.
3. Upload the C program to STM32 Nucleo-64 board.
4. Test the voting system by pressing the buttons to simulate voting.
5. Verify that the reset button clears both displays.

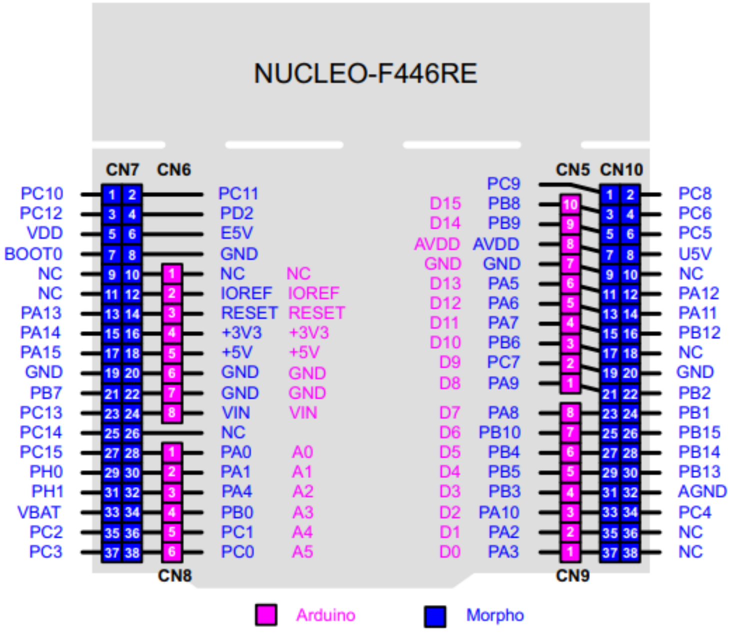

Figure 2: Nucleo 64 (F446RE) Board Pin Out

Write the are-metal C program for the voting system. Put comments in the program.

You may Cut and Paste from the program (do NOT screenshot):

Execution of Command when Button 2 pressed: Working/Not working

Execution of Command when Button 1 pressed: Working/Not working

Execution of Command when Button 0 pressed: Working/Not working

Name and Signature of Lecturer:

6 Report Writing

Title:

Objective:

Equipment/Software Used:

Procedure:

Written in third person (reporting) speech:

Result:

Fill Table and Figures for Result and attach with result.

Provide the C program with comments.

Discussion:

Discuss based on Objective and Result. There shouldn’t be any alien Objective and Result.

Conclusion:

Conclude based on Objective, Result and Discussion.